



A denoising-aided multi-task learning method for blind estimation of reverberation time

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ABSTRACT

The noise in reverberant speech severely limits the estimation accuracy of reverberation time T60 using current deep learning (DL) methods. To address this issue, this paper proposes a denoising-aided multi-task learning (DAMTL) method for blind T60 estimation. Specifically, speech denoising, as an auxiliary module, is conducted joint training with T60 estimation for more accurate prediction accuracy. These two tasks are integrated into one DL network by sharing the same encoder network, where the complex-valued spectrum is introduced to extract comprehensive high-dimensional features from noisy reverberant speech. Subsequently, complex operation of 2-D convolutional neural network (Conv2d), batch normalization and long short-term memory (LSTM) are formulated. Furthermore, the noise robustness and applicability of the DAMTL are fully discussed by comparison with state-of-the-art DL-based methods using simulated data and real-world recorded data. The results prove the effectiveness and superiority of the proposed DAMTL, especially in low signal-to-noise ratio (SNR) scenarios and practical applications.

1. Introduction

Reverberation time T60 is a basic but important room acoustic parameter that refers to the time for the sound to decay by 60 dB [1–3]. It is commonly used to evaluate the room's acoustical properties and has a significant impact on speech quality and intelligibility [4]. As a result, T60 estimation has become a critical preprocessing stage in various fields, such as dereverberation [5], speech enhancement [6], automatic speech recognition [7], and virtual reality (VR) [8].

Theoretical calculation of T60 begins with the classic Sabine's empirical formula associated with the environment geometry and absorptive characteristics [9]. Along this line of thought, subsequent improvements to this algorithm focus on the aspects of acoustic absorptivity [10], sound energy decay [11], and boundary reflection [12]. However, in a realistic environment, both the environment geometry and absorptive characteristics are not easily determined, and thus the algorithms based on Sabine's formula prove not to be very reliable. In practice, site measurement is the most common way for T60 determination using the interrupted noise method and integrated impulse response method [13]. However, this operation requires specialized equipment (e. g. omnidirectional sound source, non-directional

microphone, and data acquisition system), which is unportable, and time- and money-consuming. In response to this situation, the blind T60 estimation was born on demand as a feasible solution, which means T60 estimation can be based solely on recorded sound such as speech without heavy specialized equipment, or the prior knowledge of sound sources and room geometry. This is based on the fact that reverberant speech is a linear convolution of a clean speech signal with RIR [14], and the latter contains much information on room's acoustical properties including of course the T60.

Over the past two decades, many state-of-the-art audio signal processing algorithms have been proposed for blind T60 estimation [15]. These methods are essentially based on the statistical analysis of reverberant sound, and can be divided into two categories: model-based approaches and feature mapping-based approaches. The former assume that signal decay can be described by a statistical model and there is a certain relationship between the decay rate and reverberation time [16–18]. However, these algorithms usually give an open solution that has to be calculated by numerical optimization [19]. The latter, on the other hand, derive T60 from closely related features calculated with reverberant speech changes [20,21]. Although they contribute to somewhat improving the T60 estimation accuracy, yet usually comprise

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multiple processing blocks, requiring very specialized and sophisticated signal processing techniques [22]. In a word, efficient and accurate estimation of T60 with more advanced techniques is desired.

Recently, DL algorithm has achieved feasible advances in T60 estimation [23–27]. Unlike traditional model-based or feature mapping-based methods, the DL-based method offers a data-driven framework involving multiple constructed neural networks to estimate T60 directly from reverberant speech. Due to the excellent nonlinear mapping capability of DL, several representative studies have shown promising results. For example, Gamper et al. presented a classical convolutional neural network (CNN) approach for single-channel T60 estimation, which is conceptually straightforward and computationally efficient [28]. Xiong et al. proposed a blind room parameter estimator based on a CNN transforming the T60 estimation problem into a classification problem using simulated reverberant speech [29]. One bright point of this work is that the features from time and frequency domains of recorded speech are integrated into account as network input, resulting in promising performance. In contrast, Deng et al. used a convolutional recurrent neural network for online T60 estimation with the advantage of processing time sequences of variable length [30]. For multiple speech recordings, Srivastava et al. presented a CNN-based room parameter estimator considering both single- and inter-channel cues [31]. In a more recent work, Duangpummet et al. investigated a novel stochastic RIR that is incorporated into a CNN model for the estimation of multiple room parameters, such as T60, early decay time (EDT), clarity and speech transmission index (STI) [32]. In changing acoustic conditions, Götz et al. used three different convolutional recurrent neural networks to solve the tasks of single-band, multi-band, and multi-task parameter estimation [33]. Saini et al. employed MobileViTV3 blocks to estimate T60 and Clarity (C50), and this model can further be deployed on a mobile device and operate in a real-time environment [34].

Even though the above studies have shown promising T60 estimation results, the following challenges still exist which motivate this study: (1) It is very challenging to separate noise-like reverberation tails from actual noise, especially in low SNR scenarios. This leads to inaccurate T60 estimation and makes it difficult to use in practice [35]. (2) There have been very few studies on the applicability of DL-based T60 estimation methods considering different influencing factors. This is of great reference value for the evaluation and optimization of T60 estimation methods. (3) Most studies have been investigated only using synthetic datasets rather than real-world datasets. However, the real reverberation process is more complicated than the hypothetical linear time-invariable systems used in synthetic datasets.

To overcome the abovementioned challenges, this paper proposes a novel blind T60 estimation method using a DAMTL network. Motivated by the work that uses speech denoising as a pre-processing step before the final T60 estimation [4], this work proposes an integrated multi-task DL model where speech denoising is only used as an auxiliary module for more accurate T60 estimation through joint training of these two tasks. This new model can simplify the training process and reduce error accumulation, achieving a real end-to-end process with high computational efficiency. Specifically, the complex-valued spectrogram is introduced in the shared network of denoising and T60 estimation tasks to extract comprehensive high-dimensional features from noisy reverberant speech. Moreover, complex Conv2d, complex batch normalization and complex LSTM are formulated based on the complex operation mechanism. Next, the noise robustness and applicability of DAMTL are fully discussed by comparing with state-of-the-art DL-based methods using simulated data considering different influencing factors. To further verify the estimation performance of the proposed DAMTL in a real-world scenario, field experiments regarding five types of classrooms are finally conducted. The simulation and experiment results suggest that the DAMTL is far superior to state-of-the-art DL-based methods, which makes it to be a very promising tool for a fast and effective T60 estimation.

The contributions of this paper are threefold. (1) A new blind T60 estimation method using a denoising-aided multi-task learning network is first proposed where speech denoising is used as an auxiliary module for more accurate T60 estimation; (2) The noise-robustness and applicability of the proposed model are comprehensively investigated by discussing the effects of different experimental scenarios on T60 estimation accuracy; (3) Audio datasets with unknown RIRs and noise recorded in real-world classrooms are used to further validate the effectiveness and practicality of the proposed method.

2. Theoretical background

2.1. Acoustic model

The reverberant signal $x(t)$ can be characterized by the RIR $h(t)$ function related to room geometry, the reflection coefficients, and the distance between the sound source and receiving position. In the time domain, the $x(t)$ can be modeled as [36]

$$x(t) = s(t) * h(t) + n(t) \quad (1)$$

where $s(t)$ and $n(t)$ are the clean speech and additive noise, respectively. The operator $*$ denotes the convolution operation.

2.2. Traditional T60 estimation methods

2.2.1. Empirical calculation methods

The T60 is typically calculated using Sabine's empirical formula, which is given by

$$T60 = \frac{0.161V}{\alpha S} \quad (2)$$

where V is the volume of space; α is the average sound absorption coefficient from 0 to 1; S is the total area of floors, walls and ceilings. This formula assumes that the sound is absorbed uniformly across all surfaces in the space, and the absorption is frequency-independent.

When the average sound absorption coefficient α is greater than 0.25, the Sabine formula results in an overestimation of reverberation time. Eyring [10] gives an improved formula in terms of a large absorption coefficient, which is

$$T60 = \frac{0.161V}{-S \ln(1 - \alpha)} \quad (3)$$

2.2.2. Model-based methods

Polack [37] proposed a time domain model describing the RIR as a non-stationary stochastic process

$$h(t) = b(t)e^{-\delta t}, \quad t \geq 0 \quad (4)$$

where $b(t)$ is a random signal, typically modeled as a stationary independent and identically distributed (i.i.d) white Gaussian sequence, and δ determines the damping constant. The T60 can be then given by

$$T60 = \frac{3 \ln 10}{\delta} \quad (5)$$

This relation is only valid when the sound field is diffuse, and thus the distance between the sound source and the microphone must be greater than the critical distance. The decay model can then be defined using Eq. (4) as

$$E\{|h(t)|^2\} = \sigma_b^2 e^{-2\delta t} = \sigma_b^2 e^{\lambda_h t} \quad (6)$$

where σ_b^2 indicates the variance of $b(t)$, and λ_h is the decay rate equal to -2δ . The frequency-dependent decay rate is further given by

$$\tilde{H}(t, f) = P(f)e^{\lambda_h(f)t}, \quad t \geq 0 \quad (7)$$

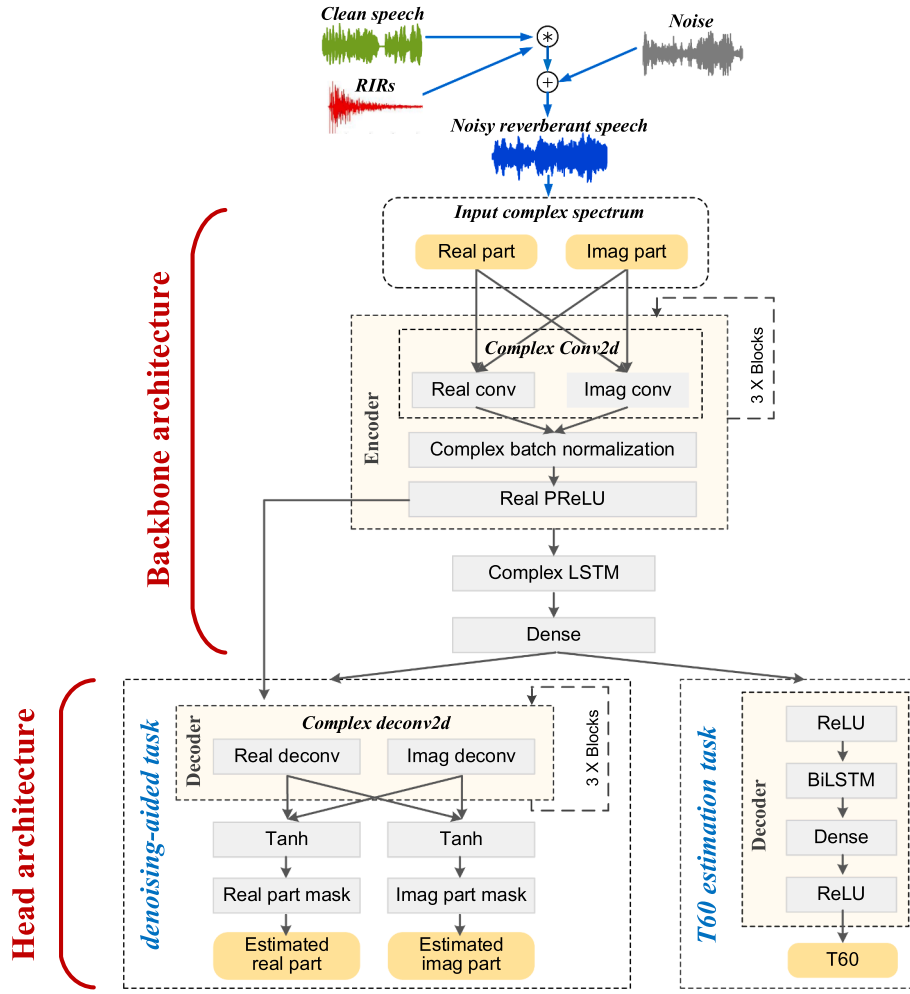


Fig. 1. Illustration of the proposed DAMTL network.

where $\tilde{H}(t, f)$ is the RIR energy envelope at time t and frequency f ; $\lambda_h(f)$ is the decay rate at frequency f ; $P(f)$ is the initial power spectral density. The logarithm formula of Eq. (7) can be written as

$$\ln \tilde{H}(t, f) = \ln P(f) + \lambda_h(f)t, t \geq 0 \quad (8)$$

The decay rate $\lambda_h(f)$ can be then estimated by fitting a line to the natural logarithm of the time–frequency energy envelope of the RIR.

2.2.3. Feature mapping-based methods

The above-discussed model-based approaches estimate T60 by determining the decay rate ρ , while the feature mapping-based methods derive T60 from a distinct feature changing with T60 from the reverberant speech. The T60 is then obtained by a learned mapping function between the extracted feature and T60. For example, Wu et al. found that the pitch strength of reverberant speech is related to reverberation degree based on the observation that reverberation can change harmonic structure [38]; Wen et al. observed that the negative-side variance of the decay rate distribution can indicate the true decay rate well, and a second-order polynomial function is used for decay rate determination given the negative-side variance [20]; Eaton et al. reduced the estimator complexity using Mel frequency bands instead of STFT frequency bands [39].

3. Proposed method

This section provides details on the proposed DAMTL network using single-channel noisy reverberant speech. As seen in Fig. 1, the whole

network includes two parts, backbone architecture and head architecture. The backbone architecture consists of shared convolution layers where both real and imaginary parts of the complex STFT spectrum are fed. Complex Conv2d, complex batch normalization and complex LSTM are further formulated based on complex-valued operation. This architecture is to extract feature representations from a complex STFT spectrum of noisy reverberant speech as input into the following head architecture.

The head architecture includes denoising-aided task and T60 estimation task. Specifically, in the denoising-aided task, real and imaginary deconvolution layers are used to downsample the extracted features, and then real and imaginary parts are estimated through their masks. In the T60 estimation task, the estimated T60 is given by the layers of rectified linear unit (ReLU), bidirectional LSTM (BiLSTM), and dense. Note that the denoising-aided task is not specifically for noise reduction, and it is just an auxiliary task that helps to improve T60 estimation accuracy, so reconstructing time-domain clean speech is unnecessary.

3.1. Backbone architecture

The core structure of this part is the encoder Conv2d block, which consists of complex Conv2d followed by complex batch normalization and real-valued parametric rectified linear unit (PReLU). Subsequently, complex LSTM and dense layer are added. Skip-connection helps with gradient flow by concentrating the encoder and decoder. The complex modules can simulate the complex-valued operation and further model the correlation between magnitude and phase [40], which will be

discussed in detail below.

3.1.1. Complex Conv2d

The designed complex Conv2d repeats three traditional Conv2d operations, allowing complex information to flow through the encoder. Define a complex-valued convolutional filter $W = A + iB$, where the real-valued matrices A and B represent the real and imaginary parts of this complex convolution kernel. Given an input complex matrix $h = x + iy$, the complex output Y is derived by the complex convolution operation.

$$Y = W \otimes h = (A^*x - B^*y) + i(B^*x + A^*y) \quad (9)$$

3.1.2. Complex batch normalization

The complex batch normalization can be viewed as one of whitening 2D vectors, scaling the complex data by the square root of their variances along each of the components [41]. This can be calculated by multiplying the inverse square root of a 2×2 covariance matrix V by a zero-centered matrix, which is

$$\tilde{x} = V^{-\frac{1}{2}}(x - E(x)) \quad (10)$$

where the V can be given as

$$V = \begin{pmatrix} V_{rr} & V_{ri} \\ V_{ir} & V_{ii} \end{pmatrix} = \begin{pmatrix} \text{Cov}(x_r, x_r) & \text{Cov}(x_r, x_i) \\ \text{Cov}(x_i, x_r) & \text{Cov}(x_i, x_i) \end{pmatrix} \quad (11)$$

This operation ensures that the $\tilde{x}$ has the standard complex distribution with mean of 0 and covariance of 1. Similar to the real-valued batch normalization calculation, the shift parameter β and scaling parameter γ are then introduced. The γ scales the input along desired new principal components to achieve a desired variance, and it can be defined by

$$\gamma = \begin{pmatrix} \gamma_{rr} & \gamma_{ri} \\ \gamma_{ir} & \gamma_{ii} \end{pmatrix} \quad (12)$$

Finally, the complex batch normalization is then given as

$$BN(\tilde{x}) = \gamma\tilde{x} + \beta \quad (13)$$

3.1.3. Complex LSTM

Similar to the derivation of complex Conv2d, the complex LSTM output Y_{out} can be given as

$$Y_{out} = (LSTM_r(X_r) - LSTM_i(X_i)) + i(LSTM_r(X_i) + LSTM_i(X_r)) \quad (14)$$

where X_r and X_i are the real and imaginary parts of complex input; $LSTM_r$ and $LSTM_i$ represent two traditional LSTMs of real and imaginary parts; $LSTM_r(X_r)$ is calculated by input X_r with $LSTM_r$, and other expressions have similar meanings.

3.1.4. Real-valued PReLU

The PReLU is used as the activation function [42]. Compared to traditional ReLU, it can learn adaptively the rectifier parameters and improve computational accuracy, which is defined as

$$PReLU(x) = \begin{cases} x, & \text{if } x > 0 \\ ax, & \text{if } x \leq 0 \end{cases} \quad (15)$$

When $a = 0$, it becomes ReLU; when a is a learnable parameter, it refers to PReLU.

3.2. Head architecture

3.2.1. Denoising-aided task

It has been shown by Zheng et al. [4] that the less noise involved in reverberant speech, the higher T60 estimation accuracy. Based on this understanding, the idea of noise-assisted evaluation is presented by introducing a denoising module as an auxiliary task. This can be done by

Table 1

Parameter description in the image method.

Parameter	Description	Value range
c	Sound velocity (m/s)	340
f_s	Sampling frequency (Hz)	16,000
s	Vector specifying the room dimensions (x, y, z) (m)	x: [3.22, 9.29] y: [4.83, 13.60] z: [2.63, 3.18]
s'	Vector specifying the (x, y, z) coordinates of the source (m)	x: [1.25, 4.07] y: [0.61, 6.03] z: 1.19
r	Distance between receiver and source (m)	[0.7, 3.5]
α	T60 (s)	[0.3, 1.5]
β	Microphone and sound source angle ($^\circ$)	[-45, 45]

introducing a weighting coefficient between denoising module and T60 estimation module. In the denoising-aided task, the decoder consists of three deconv2d blocks, aiming to transform the high-level features into complex information. The estimated real and imaginary parts will then be given based on the complex information using the masking technique [43].

3.2.2. T60 estimation task

The T60 estimation is the main purpose of the whole proposed architecture. It works well based on two aspects: (1) the effective feature extraction using complex information of the noisy reverberant speech; (2) the well-constructed denoising-aided module that offers important noise reference for T60 estimation. In this task, the BiLSTM is used to model the temporal dependencies between time steps of time series or sequence data, taking full advantage of speech features in both forward and backward directions [44]. These dependencies can be useful when the structure learns from the reverberant information at each time step.

3.3. Loss functions

The loss of denoising task L_{noise} including the real and imaginary parts of noisy speech is given by

$$L_{noise} = \frac{1}{N} \sum_i \left(\left(|\widehat{N}_{real}|_i - |N_{real}|_i \right)^2 + \left(|\widehat{N}_{imag}|_i - |N_{imag}|_i \right)^2 \right) \quad (16)$$

where $\widehat{N}_{real}$ and $\widehat{N}_{imag}$ are the estimated spectral magnitude of real and imaginary parts; N_{real} and N_{imag} are the true spectral magnitude of real and imaginary parts. Similarly, the loss of T60 estimation task L_{T60} is given by

$$L_{T60} = \frac{1}{N} \sum_i \left(|\widehat{T}_{60}|_i - |T_{60}|_i \right)^2 \quad (17)$$

where the $\widehat{T}_{60}$ and T_{60} represent the estimated and true values, respectively. The denoising task and T60 estimation task are jointly trained, so the total loss function can be written as

$$L = L_{noise} + \lambda L_{T60} \quad (18)$$

where λ is the weighting parameter between the denoising and T60 estimation tasks. It is a variable coefficient depending on network training feedback, and its final value is 2/3.

4. Experimental validation

This section aims to verify the effectiveness and priorities of the proposed method through a series of comprehensive comparison experiments from different perspectives. RIR generation is discussed followed by dataset construction, and evaluation metrics are given at last.

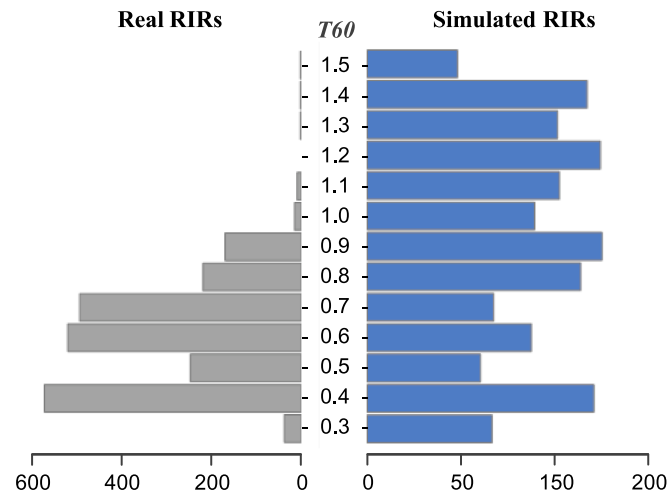


Fig. 2. Quantitative distribution of realistic and simulated RIRs with different T60s.

4.1. RIR generation

Since realistic RIRs with large T60s have a relatively low percentage in open databases, synthetic RIRs with large T60s are introduced considering the situation with long reverberation time. The synthetic RIRs are generated by the image-source method where seven rooms are considered as given in the Acoustic Characterization of Environments (ACE) challenge (Table 1). The realistic RIRs are picked from the Openair database, REVERB Challenge dataset, and RWCP database. The quantitative distribution of used RIRs is shown in Fig. 2.

4.2. Dataset construction

The experimental datasets are constructed in Table 2. For training and validation sets, the clean speech and additive noise are chosen from the public WSJ0-SI84 dataset and Deep Noise Suppression (DNS) Challenge dataset, respectively. For RIRs, refer to the previous section 4.2. To improve the noise robustness of the training model, noise is added to reverberant speech with SNRs ranging from -5 dB to 30 dB, as seen in Fig. 3. Then, the generated noisy/noiseless reverberant speech is truncated into four seconds. Following this way, there is a total of 88.89-hour speech for training purposes.

Test 1 examines the effects of different noise levels on T60 evaluation

Table 2

The description of the training, validation and test sets.

Dataset	Clean speech	RIRs	Noise	Samples with different SNRs							
				-5	0	5	10	15	20	25	30
Training	WSJ0-SI84	Synthetic and realistic	DNS challenge	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000
Validation	WSJ0-SI84	Synthetic and realistic	DNS challenge	3000	3000	3000	3000	3000	3000	3000	3000
Test 1	DNS	Synthetic and realistic	DNS challenge	1000	1000	1000	1000	1000	1000	1000	1000
Test 2	TIMIT	Synthetic or realistic	DNS challenge	1000	1000	1000	1000	1000	1000	1000	1000
Test 3	The same as test 1, but the true T60s of predicted samples range from 0.3 s to 1.5 s with an interval of 0.1 s.				6000		6000		6000		
Test 4	The same as test 1, but the speech input length is set from 0.1 s to 8 s with an interval of 0.3 s.				5400		5400			5400	
Test 5	The same to test 1, but one-third octave bands from 11.2 Hz to 8000 Hz are investigated.				14,500		14,500		14,500		
Test 6	Speech	Unknown	White noise, pink noise, babble, et al.	1000 samples with unknown RIRs and noise							
	Reading on site	Unknown	Unknown	1000 samples with unknown RIRs and noise							
	Music	Unknown	Unknown	1000 samples with unknown RIRs and noise							

performance using different methods. In this test, the synthetic and realistic RIRs are both used and all noise cases are included. Test 2 focuses on the comparison among various methods considering different RIRs. Test 3 investigates the evaluation performance using various methods considering T60 change. The true T60 values ranging from 0.3 s to 1.5 s are evenly divided into 12 intervals, and thus each interval is 0.1 s. Each sub-test set corresponding to a certain interval contains 1500 samples where different SNRs of 0 dB, 10 dB and 20 dB are considered, and a total number of 18,000 samples are used for testing. Test 4 discusses the scenarios influenced by speech input length only using the DAMTL method. The speech input length varies from 0.1 s to 8 s with an interval of 0.3 s, and thus 27 types of test data are generated. Each type includes 600 samples where different SNRs of 0 dB, 10 dB and 20 dB are considered. Thus, a total of 16,200 samples are used for testing. Test 5 discusses the effects of different subbands on T60 estimation performance. Test 6 is an actual field test where the noisy reverberant sound is recorded in actual rooms. The speech and music are utilized as the clean speech in this test.

4.3. Evaluation metrics

4.3.1. Mean absolute error (MAE)

For estimated T60 values, the first evaluation index coming to mind is MAE which gives overall errors on the test set. It can be expressed as the following equation, where $T'_{60,i}$ is the prediction value and $T_{60,i}$ is the true value.

$$MAE = \frac{\sum_{i=1}^n |T'_{60,i} - T_{60,i}|}{n} \quad (19)$$

4.3.2. Root mean square error (RMSE)

The RMSE indicates the degree of spread or distribution of the data, which can better reflect the distribution characteristics of the predicted value errors, as below.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (T'_{60,i} - T_{60,i})^2} \quad (20)$$

4.3.3. Pearson correlation coefficient (PCC)

The PCC offers a supplementary evaluation index that measures the linear correlation between two sets of data regardless of their dimensions. Its result always has a value between -1 and 1 , and the algorithm is considered better when the PCC value is closer to 1 . Its formulation can be written as

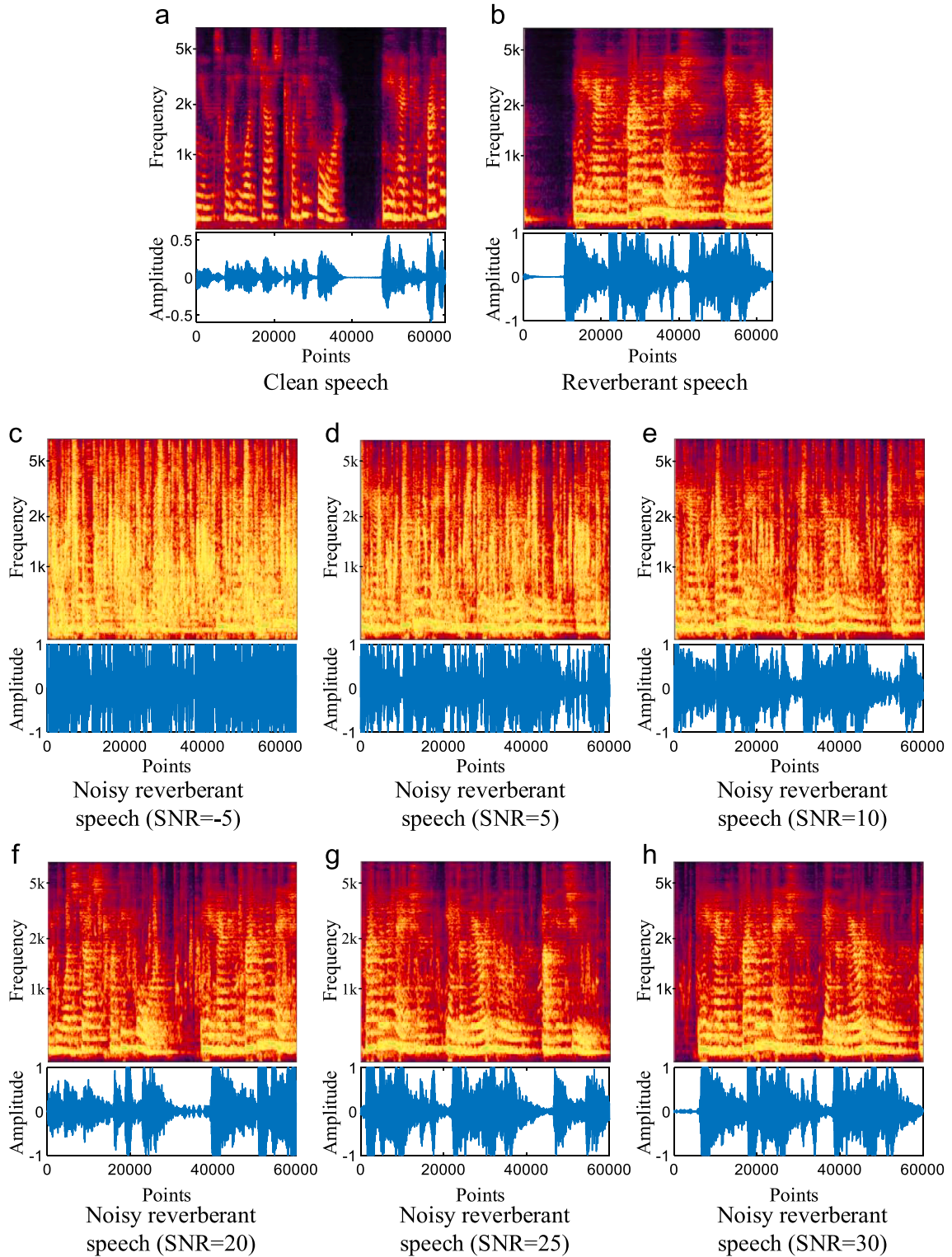


Fig. 3. The clean speech, reverberant speech, and noisy reverberant speech with different SNRs.

$$PCC = \frac{\sum_{i=1}^n (T_{60,i} - \bar{T}_{60,i}) (T'_{60,i} - \bar{T}'_{60,i})}{\sqrt{\sum_{i=1}^n (T_{60,i} - \bar{T}_{60,i})^2} \sqrt{\sum_{i=1}^n (T'_{60,i} - \bar{T}'_{60,i})^2}} \quad (21)$$

where $\bar{T}'_{60,i}$ and $\bar{T}_{60,i}$ denote the mean value of estimated $T'_{60,i}$ and its true value, respectively.

5. Results and discussion

This section discusses the T60 estimation performance of the presented DAMTL method from different perspectives corresponding to tests 1–6 in Table 2. Four state-of-the-art DL-based methods namely CNN, RE-NET, masking, and noise-aware have been introduced in [4], and they are used for comparative experiments in this paper. The first two methods only contain one stage, which directly estimate T60 using

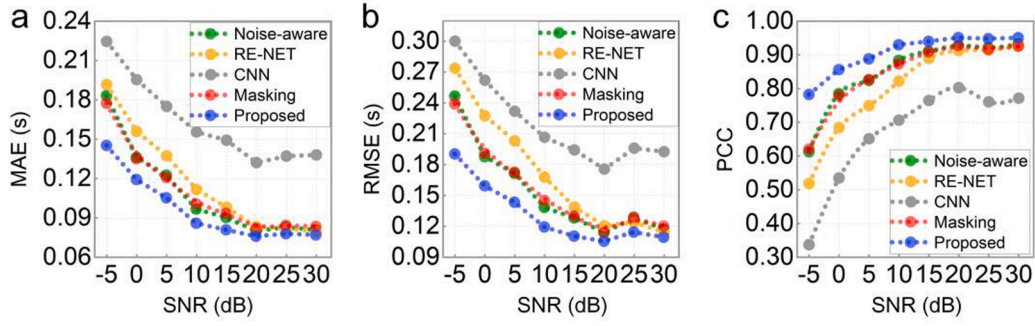


Fig. 4. The T60 estimation performance of various methods with different SNRs.

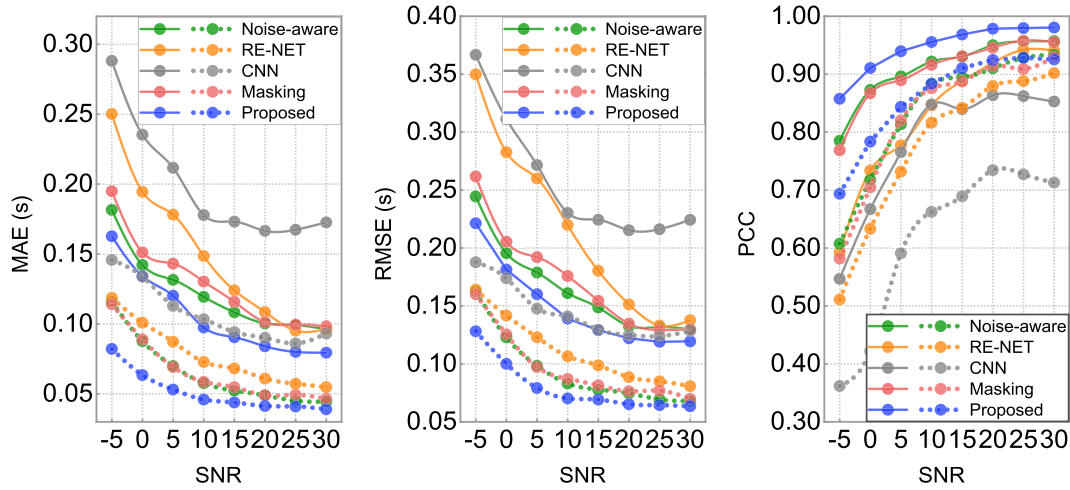


Fig. 5. Performance comparison of different methods using realistic RIRs (dotted lines) and synthetic RIRs (solid lines).

noisy reverberant speech. The remaining two methods contain two separate stages including denoising and T60 estimation procedures. For more details on these methods, please refer to [4].

5.1. Effects of different SNRs

The first investigation that comes to mind is the noise effect on the T60 estimation accuracy using the proposed DAMTL method. For this purpose, experiments using test data with different SNRs are conducted (see test 1 in Table 2) and meanwhile, the abovementioned four contrastive methods are used. Comparison results are given in Fig. 4, which show that, on the whole, the trends of MAE and RMSE are positively related and largely consistent, while they are negatively correlated with the PCC. These three metrics generally indicate that noise has a significant impact on T60 evaluation performance because the estimation accuracy gets higher as the SNR increases.

Specifically, the proposed DAMTL method obviously performs best with the minimum MAE and RMSE and the maximum PCC. The CNN method performs worst and RE-NET comes second, because these two one-stage methods directly use noisy reverberant speech for training without noise reduction. The two-stage methods do not show satisfactory results either, but they are better than the one-stage ones. The above results suggest that: (1) the front-end speech noise reduction may be an essential phase if good performance is desired; (2) the proposed method has good noise reduction ability compared to other methods.

Further, the effect of SNR range is investigated. For the SNRs ranging from -5 dB to 20 dB, the curve is steep regardless of evaluation metrics, while they flatten out when the SNR is larger than 20 dB. This means the low-SNR noise severely limits the prediction accuracy. For the low-SNR range, the curve trend of the proposed method is relatively flat

compared to other methods, which indicates it has strong robustness to noise, especially in low-SNR scenarios.

5.2. Effects of RIR types

Due to the introduction of synthetic RIRs in dataset construction, the effects of RIR types on estimation accuracy also deserve attention. Hence, experiments on test 2 are conducted and the comparison results associated with realistic and synthetic RIRs are given in Fig. 5. It is observed that most of the dotted lines are below the solid lines, which suggests that the performance using realistic RIRs generally achieves better. The good performance is mainly attributed to the fact that realistic RIRs have finer structures and more envelope information.

The image-source method used for generating synthetic RIRs allows the sound behavior calculation based on ray propagation assumptions which consider traveling directly from the source to the receiver but also indirectly mirror-like reflection. However, the actual reflective surfaces are not completely smooth, and there is also diffuse scattering on the rough reflective surface besides specular reflection. Another disadvantage of this approximation method is that the number of sound travel paths is assumed to be finite, but it turns out not. Another finding is that the DAMTL method can always get the preferable results regardless of RIR type, and this matches the conclusions drawn from the previous section well.

5.3. Effects of reverberation time

Besides investigating the effects of SNRs and RIR types, the performance regarding varying reverberant time on test 3 is also discussed. In this test, the SNRs of -5 dB, 10 dB and 20 dB are introduced. The

Table 3
Results of different methods considering various T60s.

Metrics	Methods				
	Proposed	CNN	RE-NET	Masking	Noise-aware
MAE	0.085	0.188	0.108	0.136	0.097
RMSE	0.114	0.226	0.155	0.140	0.134
PCC	0.230	0.101	0.189	0.169	0.178

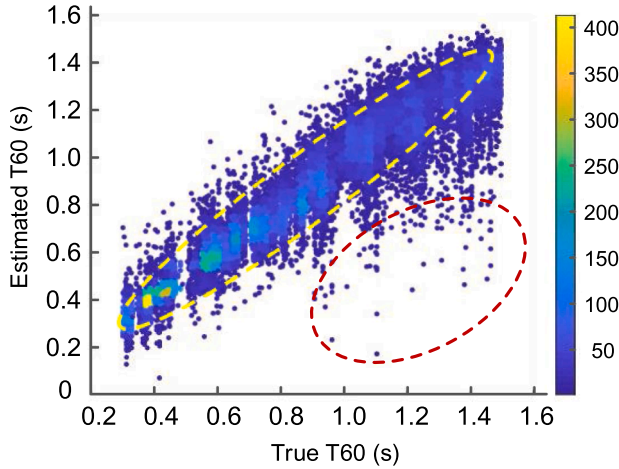


Fig. 6. Hot map of the distribution of estimated T60 values.

proposed DAMTL method performs best from the overall results of various methods, as listed in Table 3. Moreover, a simple and visual hot map is given to map the distribution of estimated T60 values, as in Fig. 6. Obviously, most points locate within the yellow dotted line along the diagonal direction, suggesting that the overwhelming majority of test samples can be predicted effectively. In addition, the DAMTL method shows poor robustness when the true T60 ranges from 0.9 s to 1.5 s, because the abnormal values within the red dotted line deviate most from their main distribution. Generally, the proposed DAMTL method yields good results, especially in the small T60 range (0.3–0.8 s), and this well meets the T60 measurement requirement of most rooms.

For further information, a box plot as well as its normal distribution of estimated points with an interval of 0.1 s is given in Fig. 7. Similar to the general trend in Fig. 6, the mean of estimated T60 values in each box

has a small fluctuation around its true T60 value. But for the T60s from 0.9 s to 1.5 s, data distribution is more dispersive compared with those of smaller T60s. The MAE and RMSE lines also reflect the fact that the larger T60 is, the relatively poorer the performance is. However, the PCC line is quite abnormal, this is because it is calculated based on an interval rather than the single T60 value. In other words, the values in each interval may be discontinuous, and their distribution change along with intervals causes a drastic PCC change.

5.4. Effects of speech input length

The speech input length can also affect T60 estimation performance. Too short speech may cause poor performance, while too long speech will lead to an overly complex prediction model consuming lots of time and computing resources. Hence, looking for a better balance between computational efficiency and estimation accuracy is necessary. Fig. 8 gives the results on speech input length using the DAMTL method according to test 4. It is shown that three metric curves become stable when the input length is larger than 1.9 s, and the estimation performance does not get better as it increases. The used input length for this work is 4 s, so it can fully meet experimental requirements from this conclusion.

5.5. Effects of different frequency bands

T60 estimation considering different subbands allows for a more detailed and accurate estimate of the reverberation characteristics of a space. This can be useful in applications such as sound design, acoustic modeling, and architectural acoustics. The DAMTL performance in a full band and 29 subband on test 5 is investigated. Box plots of the estimation error, and MAE, RMSE and PCC (right y-axis) are also illustrated. Results for one-third octave bands with SNRs of −5 dB, 10 dB and 20 dB are shown in Fig. 9. Each type of noise corresponds to 500 samples and there is a total of 1500 samples in terms of an individual band. Fig. 9 shows that the trends of MAE and RMSE are generally consistent. Combined with the PCC trend, it can be seen that performance improves significantly from 1120 to 5600 Hz. Especially for 1400 to 2800 Hz, it achieves the best. The performance of overall frequency dependence also reflects the algorithm design, which is based on speech-specific stimulation with high speech energy, i.e., 1–3 k Hz.

5.6. Experiments using recorded room data

To further verify the practical ability of the proposed DAMTL

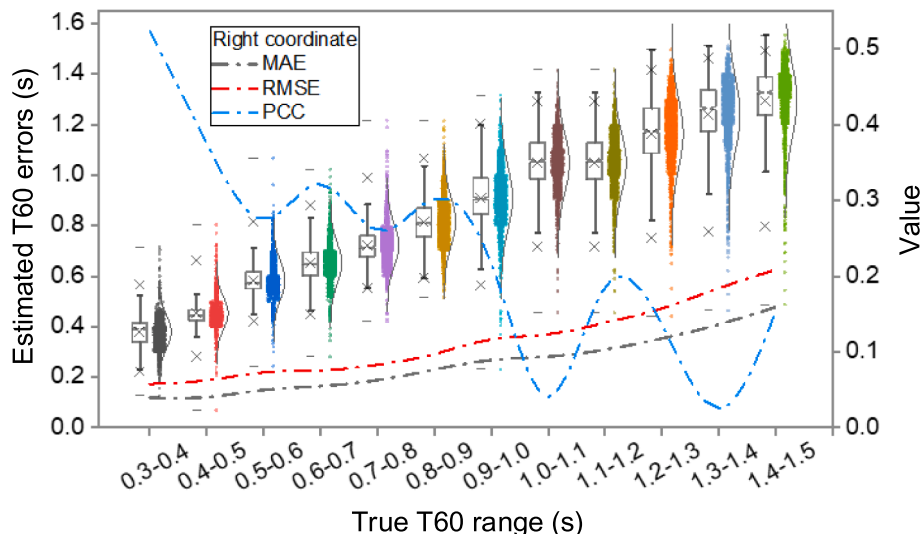


Fig. 7. Box plot as well as its normal distribution with an interval of 0.1 s.

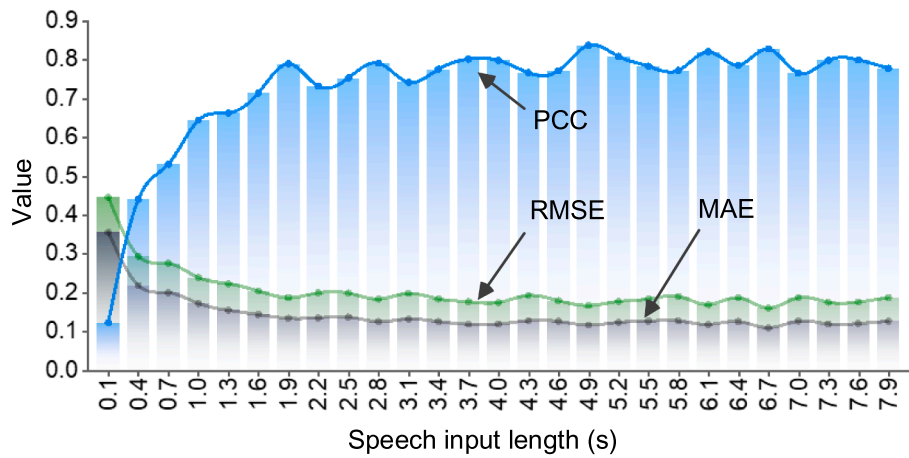


Fig. 8. Effect of different speech input lengths on T60 estimation.

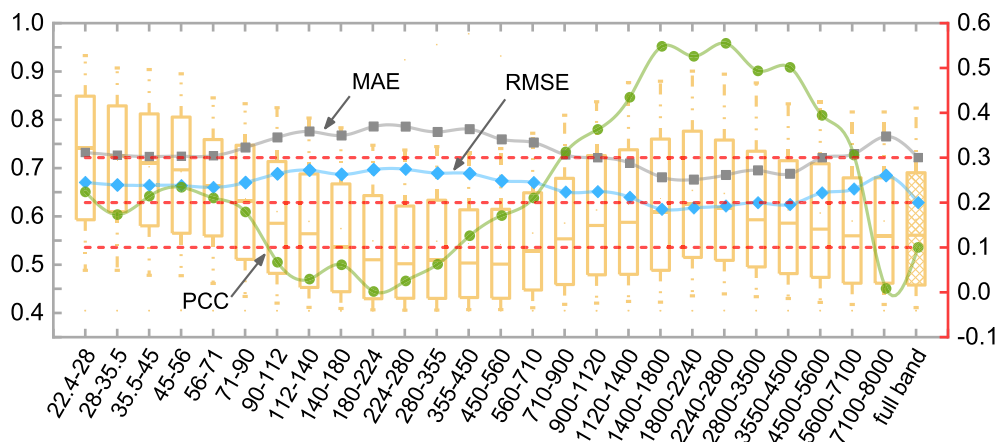


Fig. 9. Box plots of the estimation error, and MAE, RMSE and PCC (right y-axis).

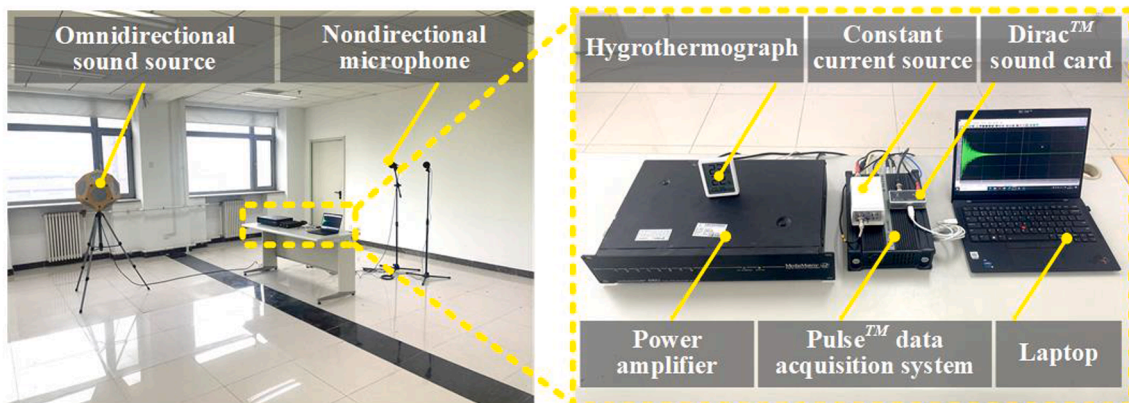


Fig. 10. The experimental equipment and its layout.

method, a series of experiments considering real-world scenarios are investigated. Two experimental systems including PulseTM and DiracTM are used to measure T60, as seen in Fig. 10, and the former is also used to record reverberant sound. The performance of the DAMTL method is then discussed by comparing the measurement T60 values with the predicted values based on recorded reverberant sound. A detailed experimental description is discussed in detail below.

5.6.1. Experimental setups

The PulseTM is a versatile, task-oriented sound and vibration analysis system, while the DiracTM is a sophisticated tool to calculate a very wide range of acoustical parameters from impulse response measurements of an enclosure. The pulse system is essentially a data acquisition device, and the dirac system contains a sound card and a constant current source. The sound source and microphone are shared for both systems. In addition, the hygrothermograph is used to measure both temperature and relative humidity, because these parameters can affect the speed of



Fig. 11. The measured rooms with different categories and volumes.

Table 4

The description of measured rooms.

Categories	Room	Size	Capacity (seating)	Volume (m^3)
Common classroom	Room 1	Small	40	210
Common classroom	Room 2	Medium	48	270
Smart classroom	Room 3	Medium	48	283
Terrace classroom	Room 4	Large	154	772
Lecture classroom	Room 5	Large	302	1656

sound. For the pulse system, multiple measurements are needed based on the interrupted noise method. But for the dirac system using integrated impulse response method, one experiment can meet the accuracy requirement.

5.6.2. Experimental procedures

(a) T60 measurement

As seen in Fig. 11 and Table 4, five real rooms with different categories and volumes are considered in the experiments, which cover

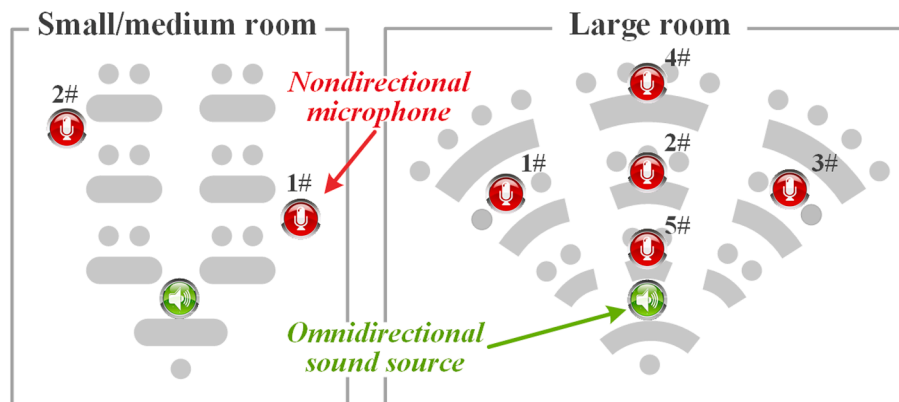


Fig. 12. Measurement point layout in different rooms.



Fig. 13. SPL measurement of clean speech and noise at one position.

Table 5
The T60s measured by different systems.

Positions	T60s measured by pulse/dirac				
	Room 1	Room 2	Room 3	Room 4	Room 5
1#	1.03/1.06	1.20/1.25	0.65/0.65	1.15/1.17	1.73/0.78
2#	1.07/1.06	1.35/1.32	0.78/0.75	1.21/1.23	1.90/0.81
3#	-/-	-/-	-/-	1.43/1.38	0.72/0.75
4#	-/-	-/-	-/-	1.35/1.36	0.86/0.88
5#	-/-	-/-	-/-	1.50/1.53	0.88/0.90
Average	1.05/1.06	1.28/1.29	0.72/0.70	1.33/1.34	0.82/0.83

common classrooms, smart rooms, terrace classrooms and lecture classroom. Single-channel multiple measurements are conducted by both systems, and the measurement point placement is shown in Fig. 12. Two and five positions are arranged for small/medium and large rooms, respectively.

(b) Sound recording

For sound recording, the recording locations are the same as the T60 measurement points, but the location and type of sound source are not exactly the same. The audio sample rate is 48 k.

Scenario 1: Four dry speeches read by men and women in Chinese and English, are played by the omnidirectional sound source at the podium position as seen in Fig. 12. Each speech lasts for about a minute. Furthermore, different levels of noise severity are also taken into consideration. The noise type includes white noise, pink noise, babble, and environmental noise such as instrumental sound, steam whistle, birdcall and so on. To obtain the reverberation speech with a certain SNR, the noise and clean speech are measured separately at the same location: first, adjust the volume of noise and clean speech to meet the desired sound pressure level (SPL) at the specified location (see Fig. 13); secondly, record reverberant sound when playing both noise and clean speech. Thus, the noisy reverberant sound with SNRs of -5 , 0 and 10 is obtained, and each sound is about 4 min.

Scenario 2: A male talker stands in the front of the room, reading a Chinese textbook for about two minutes. Then, he walks around the room and keeps reading for two minutes again. Background noise is the real ambient noise.

Scenario 3: It is the same as Scenario 2, but the reading by man has been changed to playing music with a hand-held stereo.

5.6.3. Experimental results

(a) T60 measurement

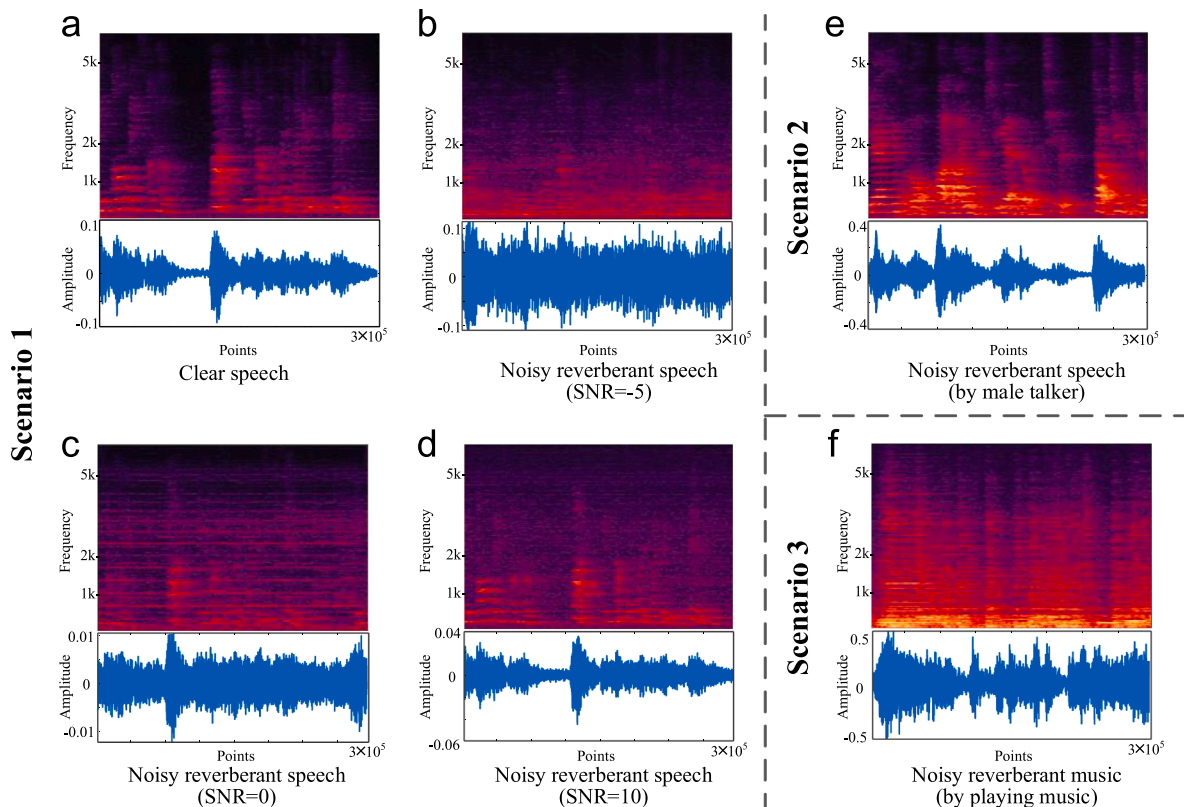


Fig. 14. The recorded speech/music with different SNRs.

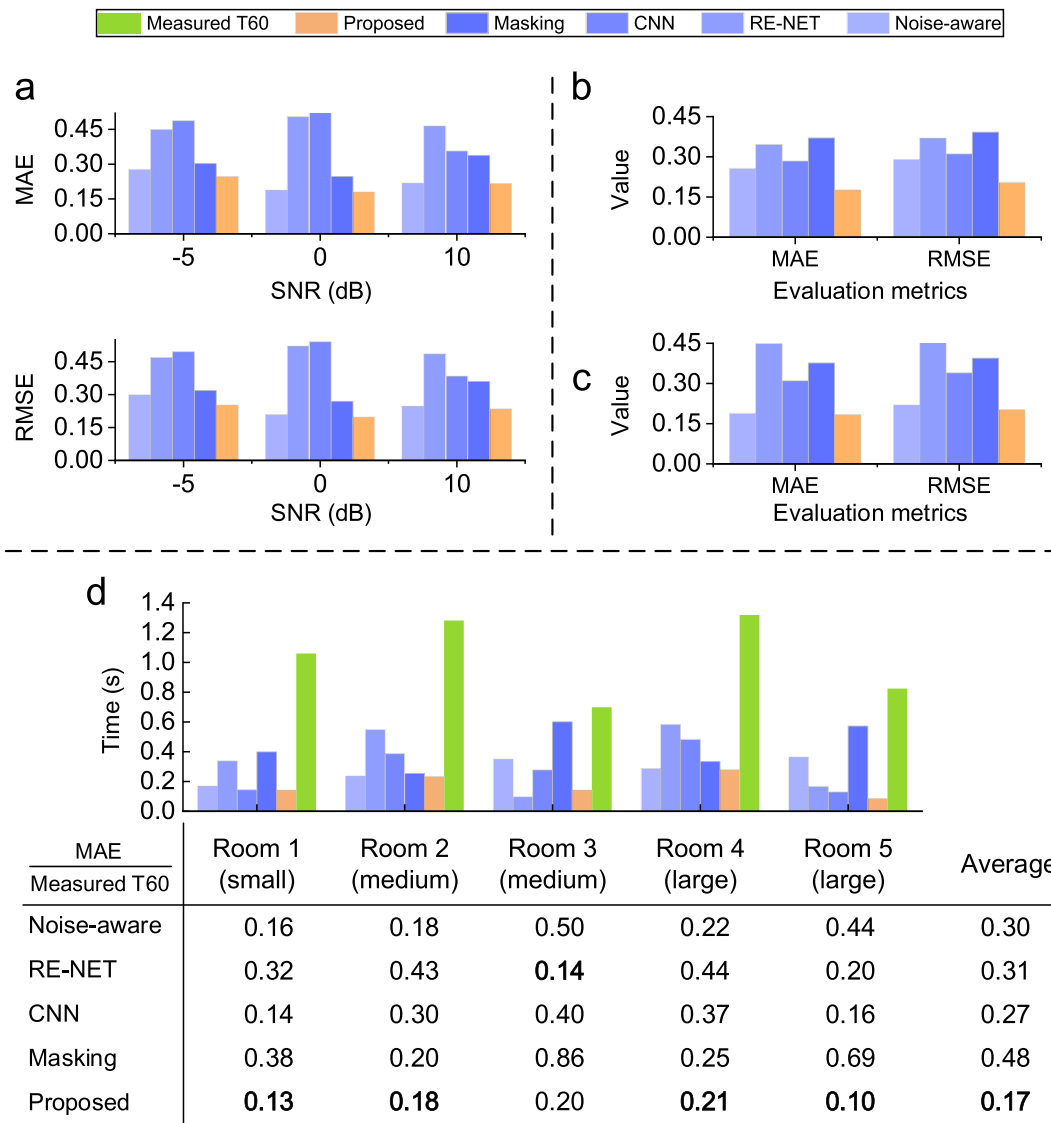


Fig. 15. T60 performance using real-world recorded sound.

Three T60 measurements in one position using the pulse system are conducted, and the mean measurement value is listed in Table 5. However, only one experiment is adequate for the dirac system. According to Tables 4 and 5, Room 3 with the smallest T60 gets the best sound quality, because the smart room is surrounded by soundproof cotton, and pays more attention to multimedia sound interaction. It is also observed the T60 of the smallest Room 1 is relatively large, implying that small rooms can even get a very poor sound quality. Another finding is that the auditory quality in Room 5 is better than Room 4 even though its volume is larger. This is due to its unique fan shape, which suppresses too much reverberation. Also, it is noted that these measured T60s have a small error regardless of measurement methods.

(b) Sound recording

For more detailed information, the spectrogram and waveform of the selected recorded sound at position 1# in Room 4 (large room) are given. For scenario 1, the speech is read by a female in English, and the babble, environmental noise and white noise are added to reverberant speech to generate the noisy reverberant speech with SNRs of -5 dB, 0 dB and 10 dB (Fig. 14(a–d)). For scenario 2, the speech is read by a male

talker walking around the room, and the added noise is unknown environmental noise (Fig. 14(e)). For scenario 3, the sound is music played by a hand-held stereo moving around the room, and the added noise is also unknown (Fig. 14(f)).

(c) Performance with different methods

The real-world recorded sound in scenarios 1–3 is used to further verify the proposed DAMTL method. According to test 6, each case includes 1000 samples. In scenario 1, the noise with different SNRs is introduced, while in scenarios 2–3 only background noise exists and the noise intensity is unknown. Also, the four constructive methods are used. The experimental results are summarized in Fig. 15. Generally, the PCC discusses the linear correlation considering two variables. But in theory, the T60 in each room is a constant, so this index is invalid here and only MAE and RMSE are considered.

Fig. 15(a) shows the average results of five rooms using recorded speech in scenario 1. It is observed that the proposed DAMTL always gets the minimum error, regardless of evaluation metrics and noise severity. This method also seems insensitive to noise, because the numerical fluctuation of MAE and RMSE is not very large with the change of SNR. It can also be concluded that the two-stage methods outperform the one-

stage methods.

Fig. 15(b–c) gives the mean errors in scenarios 2 and 3. The obvious difference between these two figures is the errors when playing music are larger, since that the musical background is inherently mixed with reverberation. From this conclusion, playing music to generate reverberation sound is unreliable. Similar to the conclusions drawn from Fig. 15(a), the DAMTL performs best.

Fig. 15(d) illustrates the results regarding different rooms, which aims to discuss the T60 prediction ability in terms of different reverberation time. The ultimate goal of this work is to use the proposed DAMTL for actual T60 estimation, so the performance in real scenes deserves further investigation. Focusing solely on absolute errors is sometimes meaningless, since for rooms with a large T60, it may be relatively large even though the used method performs well. Hence, a relative index equaling MAE divided by the measured T60, is given to make a more objective assessment, which is calculated in the table below. From the last column in the table, the average error is 0.17, and this makes it available to use in practice.

6. Conclusions

In this work, a novel blind T60 estimation method using a denoising-aided multi-task learning network is proposed, where speech denoising is presented as an auxiliary module to conduct joint training with T60 estimation for more accurate T60 prediction. To extract comprehensive high-dimensional features, the complex-valued spectrum of reverberant speech is introduced. Moreover, complex Conv2d, batch normalization, and LSTM are formulated. Simulation and experiment results prove the effectiveness and superiority of this new method by comparing it with the state-of-the-art DL-based T60 estimation methods. Based on the results obtained, the following conclusions can be drawn:

- (1) The proposed DAMTL method exhibits strong robustness to noise, especially in low-SNR scenarios, such as SNRs are -5 dB, 0 dB and 5 dB. The good performance is due to the advanced multi-task network where denoising module contributes to improving T60 estimation accuracy.
- (2) This new method performs better in the following conditions: the realistic RIRs are considered instead of synthetic RIRs; the T60 to be estimated ranges from 0.3 s to 0.8 s; the input speech length is larger than 1.9 s; the frequency band range is from 1400 to 2800 Hz.
- (3) The newly presented approach develops a very good practical application ability. In the situation when a random speech is generated by a talker walking around a room with normal background noise, the T60 estimation error can be controlled within 0.17 .

In summary, the proposed DAMTL performs well on T60 estimation considering low-SNR scenarios. It has good applicability and is sufficient for a fast and rough estimation of room acoustic parameters. Despite its beneficial performance, the model parameters are slightly larger. Future extension work can focus on model simplification and computational optimization. From an application perspective, this model can also be used for the estimation of other acoustic parameters, such as the ED, STI and clarity.

CRediT authorship contribution statement

Yulong Zhang: Writing – original draft, Methodology, Conceptualization. **Jinqiu Sang:** Validation, Software. **Chengshi Zheng:** Formal analysis, Data curation. **Xiaodong Li:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] H. Kuttruff, E. Mommertz, Room Acoustics, Applied Science Publishers, 2013.
- [2] B.M.R. Bharathi, A.R. Mohanty, Time delay estimation in reverberant and low SNR environment by EMD based maximum likelihood method, *Measurement* 137 (2019) 655–663.
- [3] J. Olsson, A. Linderholt, K. Jarnerö, et al., Incremental use of FFT as a solution for low BT-product reverberation time measurements, *Appl. Acoust.* 203 (2023) 109191.
- [4] K. Zheng, C. Zheng, J. Sang, et al., Noise-robust blind reverberation time estimation using noise-aware time-frequency masking, *Measurement* 192 (2022) 110901.
- [5] J. Fischer, C. Doolan, An improved eigenvalue background noise reduction method for acoustic beamforming, *Mech. Syst. Sig. Process.* 140 (2020) 106702.
- [6] C. Gong, M. Wu, J. Guo, et al., Multichannel narrowband active noise control system with a frequency estimator based on DFT coefficients, *J. Sound Vib.* 521 (2022) 116660.
- [7] M. Benzeghiba, R. De Mori, O. Deroo, et al., Automatic speech recognition and speech variability: a review, *Speech Comm.* 49 (10–11) (2007) 763–786.
- [8] C. Anthes, R.J. García-Hernández, M. Wiedemann, et al., State of the art of virtual reality technology, in: 2016 IEEE Aerospace Conference, IEEE, 2016, pp. 1–19.
- [9] W.C. Sabine, C.H. Dwight, Collected papers on acoustics, *Am. J. Phys.* 34 (4) (1964) 370–371.
- [10] C.F. Eyring, Reverberation time in “dead” rooms, *J. Acoust. Soc. Am.* 1 (2A) (1930) 217–241.
- [11] H. Kuttruff, Reverberation and effective absorption in rooms with Diffuse Wall reflexions, *Acta Acust. Acust.* 35 (3) (1976) 141–153.
- [12] H. Kuttruff, A simple iteration scheme for the computation of decay constants in enclosures with diffusely reflecting boundaries, *J. Acoust. Soc. Am.* 98 (1) (1995) 288–293.
- [13] ISO 3382-2, Acoustics – Measurement of Room Acoustic Parameters – Part 2: Reverberation Time in Ordinary Rooms, International Organization for Standardization, Brussels, Belgium, 2008.
- [14] L. Čurović, J. Murovec, T. Novaković, et al., Time-frequency methods for characterization of room impulse responses and decay time measurement, *Measurement* 196 (2022) 111223.
- [15] H. Löllmann, A. Brendel, W. Kellermann, Comparative Study of Single-channel Algorithms for Blind Reverberation Time Estimation, *Universitätsbibliothek der RWTH Aachen*, 2019.
- [16] K. Lebart, J.M. Boucher, P.N. Denbigh, A new method based on spectral subtraction for speech dereverberation, *Acta Acust. Acust.* 87 (3) (2001) 359–366.
- [17] T.M. Prego, A.A. de Lima, S.L. Netto, et al., A blind algorithm for reverberation-time estimation using subband decomposition of speech signals, *J. Acoust. Soc. Am.* 131 (4) (2012) 2811–2816.
- [18] A. Hamza, T. Jan, A. Jehangir, et al., A novel approach for blind estimation of reverberation time using gamma distribution model, *J. Electr. Eng. Technol.* 11 (2) (2016) 529–536.
- [19] R. Ratnam, D.L. Jones, B.C. Wheeler, et al., Blind estimation of reverberation time, *J. Acoust. Soc. Am.* 114 (5) (2003) 2877–2892.
- [20] J.Y.C. Wen, E.A.P. Habets, P.A. Naylor, Blind estimation of reverberation time based on the distribution of signal decay rates, in: 2008 IEEE International Conference on Acoustics, Speech and Signal Processing, 2008, pp. 329–332.
- [21] T.H. Falk, W.Y. Chan, Temporal dynamics for blind measurement of room acoustical parameters, *IEEE Trans. Instrum. Meas.* 59 (4) (2010) 978–989.
- [22] H. Löllmann, E. Yilmaz, M. Jeub, et al., An improved algorithm for blind reverberation time estimation, in: Proceedings of International Workshop on Acoustic Echo and Noise Control (IWAENC), 2010, pp. 1–4.
- [23] R. Scharer, M. Vorländer, Blind reverberation time estimation. Proceedings of the International Conference on Acoustics, Sydney, Australia, 2010.
- [24] X. Xiao, S. Zhao, X. Zhong, et al., Learning to estimate reverberation time in noisy and reverberant rooms. Sixteenth Annual Conference of the International Speech Communication Association, 2015.

- [25] M. Lee, J.H. Chang, Blind estimation of reverberation time using deep neural network, in: 2016 IEEE International Conference on Network Infrastructure and Digital Content (IC-NIDC), 2016, pp. 308–311.
- [26] F. Xiong, S. Goetze, B. Kollmeier, et al., Joint estimation of reverberation time and early-to-late reverberation ratio from single-channel speech signals, *IEEE/ACM Trans. Audio Speech Lang. Process.* 27 (2) (2018) 255–267.
- [27] N.J. Bryan, Impulse response data augmentation and deep neural networks for blind room acoustic parameter estimation, in: ICASSP 2020–2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2020, pp. 1–5.
- [28] H. Gamper, I.J. Tashev, Blind reverberation time estimation using a convolutional neural network, in: 2018 16th International Workshop on Acoustic Signal Enhancement (IWAENC), IEEE, 2018, pp. 136–140.
- [29] F. Xiong, S. Goetze, B. Kollmeier, et al., Exploring auditory-inspired acoustic features for room acoustic parameter estimation from monaural speech, *IEEE/ACM Trans. Audio Speech Lang. Process.* 26 (10) (2018) 1809–1820.
- [30] S. Deng, W. Mack, E.A.P. Habets, Online blind reverberation time estimation using CRNNs, in: INTERSPEECH, 2020, pp. 5061–5065.
- [31] P. Srivastava, A. Deleforge, E. Vincent, Blind room parameter estimation using multiple multichannel speech recordings, in: 2021 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), IEEE, 2021, pp. 226–230.
- [32] S. Duangpummet, J. Karnjana, W. Kongprawechnon, et al., Blind estimation of speech transmission index and room acoustic parameters based on the extended model of room impulse response, *Appl. Acoust.* 185 (2022) 108372.
- [33] P. Götz, C. Tuna, A. Walther, et al., Online reverberation time and clarity estimation in dynamic acoustic conditions, *J. Acoust. Soc. Am.* 153 (6) (2023) 3532–3542.
- [34] S. Saini, J. Peissig, Blind room acoustic parameters estimation using mobile audio transformer, in: 2023 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), IEEE, 2023, pp. 1–5.
- [35] N. Moritz, J. Anemüller, B. Kollmeier, Amplitude modulation spectrogram based features for robust speech recognition in noisy and reverberant environments, in: 2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2011, pp. 5492–5495.
- [36] P. Filippi, A. Bergassoli, D. Habault, et al., *Acoustics: Basic Physics, Theory, and Methods*, Elsevier, 1998.
- [37] J.D. Polack, La transmission de l'énergie sonore dans les salles, Le Mans, 1988.
- [38] M. Wu, D.L. Wang, A pitch-based method for the estimation of short reverberation time, *Acta Acust. Acust.* 92 (2) (2006) 337–339.
- [39] J. Eaton, N.D. Gaubitch, P.A. Naylor, Noise-robust reverberation time estimation using spectral decay distributions with reduced computational cost, in: 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, 2013, pp. 161–165.
- [40] K. Tan, D.L. Wang, Complex spectral mapping with a convolutional recurrent network for monaural speech enhancement, in: ICASSP 2019–2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019, pp. 6865–6869.
- [41] C. Trabelsi, O. Bilaniuk, Y. Zhang, et al. Deep complex networks. *arXiv preprint arXiv:1705.09792*, 2017.
- [42] K. He, X. Zhang, S. Ren, et al., Delving deep into rectifiers: Surpassing human-level performance on imagenet classification, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2015, pp. 1026–1034.
- [43] N. Li, P.C. Loizou, Factors influencing intelligibility of ideal binary-masked speech: implications for noise reduction, *J. Acoust. Soc. Am.* 123 (3) (2008) 1673–1682.
- [44] X. Cui, Z. Chen, F. Yin, Multi-objective based multi-channel speech enhancement with BiLSTM network, *Appl. Acoust.* 177 (2021) 107927.